

Department of Chemistry, NIT Kurukshetra
B. Tech. II Semester, EE, ECE and M&V [Exam: Mid Semester – II]

Subject: Chemistry (CHIC-103)

Date: 07.04.2026

Marks: 15 (Time: 50 min)

Instructions: (a) Write down your Roll No. and Section on the top corner of the answer sheet, (b) No extra answer sheet will be provided, (c) Answer to all questions.

Question No.	Description	Marks
1	Why Zn-HgO cell was condemned over Zn-Ag ₂ O cell? Explain cell reaction (anode, cathode and overall) of Zn-Ag ₂ O with cell diagram.	3
2	Write at least three important differences between secondary battery and fuel cells.	3
3	What is working principle of the Molten Carbonate Fuel Cells? Describe complete cell reactions using cell diagram showing flow of fuels and end products.	4
4	Fill the following blanks a) The name of the electrolyte uses in the PEMFC is..... b) Cathode of the Lithium ion battery is made up of c) The general cell representation of lead acid battery is $Pb PbSO_4 H_2SO_4 PbO_2 Pb$ d) The Noble prize in Chemistry 2019 was jointly awarded to John B. Goodenough, M. Stanley Whittingham, and A. Yoshino for.... $Li-ion$ Batteries. e) Potential energy difference in the Zn/Cu system is	5